

CURRENT POSITION	Sherbrooke University , Sherbrooke, Quebec, Canada Assistant Professor, 2026–present
EDUCATION	University of Rochester , Rochester, New York Postdoctoral Fellow, 2023–2025 California Institute of Technology , Pasadena, California Ph.D. in Theoretical Chemistry, 2018–2023 University of Toronto , Toronto, Canada B.Sc. (Honours) in Chemistry with High Distinction, 2014–2018
FELLOWSHIPS AND AWARDS	Steadman Award 2024 Student Leader Award 2024 Gray-Hill Award Lecture 2023 Patricia Beckman Graduate Fellowship 2018 Michael Rebryk Memorial Scholarship 2018 Ivan Szak Scholarship in Chemistry 2018 St. Michael's College Silver Medal 2018 University of Toronto Excellence Award 2018 St. Michael's College In-Course Scholarship 2018 Canadian Society for Chemistry Silver medal 2018 CQIQC Undergraduate Summer Research Program 2017 University of Toronto Excellence Award 2017 F. E. Beamish Scholarship in Chemistry 2017 Buduchnist Credit Union Scholarship 2017 Ivan Szak Scholarship in Chemistry 2017 Michael Both Award for Outstanding Commitment to Dance 2017 John Melady Memorial Scholarship 2017 C. W. Burton In-Course Scholarship 2017 Gollop Memorial Undergraduate Scholarship in Chemistry 2017 Dean's List Scholar 2017 University of Toronto Excellence Award 2016 Kupcinet-Getz Research Scholarship 2015 University of Toronto Mississauga Honour Roll 2015 Erindale Admission Scholarship 2014 Scholarship of the President of Ukraine 2014 (<i>awarded annually to approximately 250 highest achieving high school students out of over a million</i>) Lutsk's student of the year 2014 (<i>awarded annually to the highest achieving high school graduate out of approximately 10,000</i>) First Prize at Intel-Eco Ukraine 2014, the national stage of Intel ISEF 2014 Gold medal at the International Ecology Project Olympiad 2013

PUBLICATIONS

16. Dora, R; Korol, R; Tiwari, V; Chourasiya, R; Franco*, I Light-Matter Interactions Beyond the Dipole Approximation in Extended Systems Without Multipole Expansion. *arXiv* **2026**. DOI: [10.48550/arXiv.2603.10271](https://doi.org/10.48550/arXiv.2603.10271)
15. Korol*, R; Nichol, J; Franco, I How NOT to Build Control-Target Gates in Semiconductor Quantum Dots and Beyond. *arXiv* **2026**. DOI: [10.48550/arXiv.2608.00733](https://doi.org/10.48550/arXiv.2608.00733)
14. Tiwari, V; Korol, R; Franco*, I Robust Purely Optical Signatures of Floquet States in Laser-Dressed Crystals. *Phys. Rev. Lett.* **2025**, 135 (18), 186901. DOI: [10.1103/5ywx-7db5](https://doi.org/10.1103/5ywx-7db5)
13. Korol*, R; Chen, X; Franco*, I High-Frequency Tails in Spectral Densities. *J. Phys. Chem. A* **2025**, 129 (15), 3587-3596. DOI: [10.1021/acs.jpca.5c00943](https://doi.org/10.1021/acs.jpca.5c00943)
12. Turner*, A C; Korol, R; Bill, M; Stolper, D A Stable isotope equilibria in the dihydrogen-water-methane-ethane-propane system. Part 2: Experimental determination of hydrogen isotopic equilibrium for ethane-H₂ from 30 to 200 °C and propane-H₂ from 75 to 200 °C. *Geochim. Cosmochim. Acta* **2025**, 396, 91-106. DOI: [10.1016/j.gca.2025.02.033](https://doi.org/10.1016/j.gca.2025.02.033)
11. Korol*, R; Turner, A C; Nandi, A; Bowman, J M; Goddard, W A; Stolper, D A Stable isotope equilibria in the dihydrogen-water-methane-ethane-propane system. Part 1: Path-integral calculations with CCSD(T) quality potentials. *Geochim. Cosmochim. Acta* **2025**, 396, 71-90. DOI: [10.1016/j.gca.2025.02.028](https://doi.org/10.1016/j.gca.2025.02.028)
10. Turner, A C; Korol, R; Eldridge, D L; Bill, M; Conrad, M E; Miller, T F; Stolper*, D A Experimental and theoretical determinations of hydrogen isotopic equilibrium in the system CH₄-H₂-H₂O from 3 to 200 °C. *Geochim. Cosmochim. Acta* **2021**, 314, 223-269. DOI: [10.1016/j.gca.2021.04.026](https://doi.org/10.1016/j.gca.2021.04.026)
9. Korol, R; Rosa-Raíces, J L; Bou-Rabee, N; Miller*, T F Dimension-free path-integral molecular dynamics without preconditioning. *J. Chem. Phys.* **2020**, 152 (10), 104102. DOI: [10.1063/1.5134810](https://doi.org/10.1063/1.5134810)
8. Eldridge, D L; Korol, R; Lloyd, M K; Turner, A C; Webb, M A; Miller, T F; Stolper*, D A Comparison of Experimental vs Theoretical Abundances of ¹³CH₃D and ¹²CH₂D₂ for Isotopically Equilibrated Systems from 1 to 500 °C. *ACS Earth Space Chem.* **2019**, 3 (12), 2747-2764. DOI: [10.1021/acsearthspacechem.9b00244](https://doi.org/10.1021/acsearthspacechem.9b00244)
7. Korol, R; Bou-Rabee, N; Miller*, T F Cayley modification for strongly stable path-integral and ring-polymer molecular dynamics. *J. Chem. Phys.* **2019**, 151 (12), 124103. DOI: [10.1063/1.5120282](https://doi.org/10.1063/1.5120282)
6. Korol, R; Segal*, D Machine Learning Prediction of DNA Charge Transport. *J. Phys. Chem. B* **2019**, 123 (13), 2801-2811. DOI: [10.1021/acs.jpcc.8b12557](https://doi.org/10.1021/acs.jpcc.8b12557)
5. Korol, R; Segal*, D From Exhaustive Simulations to Key Principles in DNA Nanoelectronics. *J. Phys. Chem. C* **2018**, 122 (8), 4206-4216. DOI: [10.1021/acs.jpcc.7b12744](https://doi.org/10.1021/acs.jpcc.7b12744)
4. Korol, R; Kilgour, M; Segal*, D ProbeZT: Simulation of transport coefficients of molecular electronic junctions under environmental effects using Büttiker's probes. *Comput. Phys. Commun.* **2018**, 224, 396-404. DOI: [10.1016/j.cpc.2017.10.005](https://doi.org/10.1016/j.cpc.2017.10.005)
3. Korol, R; Kilgour, M; Segal*, D Thermopower of molecular junctions: Tunneling to hopping crossover in DNA. *J. Chem. Phys.* **2016**, 145 (22), 224702. DOI: [10.1063/1.4971167](https://doi.org/10.1063/1.4971167)
2. Longobardi, L E; Zatsepin, P; Korol, R; Liu, L; Grimme, S; Stephan*, D W Reactions of Boron-Derived Radicals with Nucleophiles. *J. Am. Chem. Soc.* **2016**, 139 (1), 426-435. DOI: [10.1021/jacs.6b11190](https://doi.org/10.1021/jacs.6b11190)
1. Korol*, R V; Yanchuk, O M; Marchuk, O V; Orlov, V F; Moroz, I A; Vyshnevskiy, O A Size Stabilizers in Two-electrode Synthesis of ZnO Nanorods. *Phys. Chem. Solid St.* **2021**, 22 (2), 380-387. DOI: [10.15330/pcss.22.2.380-387](https://doi.org/10.15330/pcss.22.2.380-387)

PRESENTATIONS AND AWARDS	4 th CEMDI-PAIMS Symposium , McGill University, Montreal, Quebec	2026
	<u>Invited talk</u> : “Illuminating Materials by Computations: Characterization of Beyond-Dipole Effects and Signatures of Time Crystals” Session chair (Solid-State Ionics)	
	Canadian Chemistry Conference and Exhibition , Toronto, Ontario	2026
	<u>Invited talk</u> : “A window to Earth’s past with the help of theoretical chemistry” Contributed talk: “How NOT to build control-target gates in semiconductor quantum dots and beyond”	
	Physical, Theoretical and Computational Chemistry (PTC) General Session, Toronto, Ontario	2026
	Session chair	
	CQIQC Seminar, University of Toronto, Toronto, Ontario	2026
	<u>Invited talk</u> : “How NOT to build control-target gates in semiconductor quantum dots and beyond”	
	CERMM Seminar, Concordia University, Montreal, Quebec	2026
	<u>Invited talk</u> : “Quantum Effects in Complex Systems”	
	“Lunch and Learn” series, Department of Chemistry, Sherbrooke University, Quebec	2026
	Talk: “Quantum Effects in Complex Systems”	
	CCQS Seminar at the Physics department, University of Rochester, Rochester, New York	2025
	Talk: “How NOT to build control-target gates in semiconductor quantum dots and beyond”	
	Summer of Molecular Quantum Information Science Seminars Series	2025
	Talk: “Opening quantum dynamics on gate-defined quantum dots with RLC circuits”	
	Rochester Conference on Coherence and Quantum Science, Rochester, New York	2025
	Poster: “Hybrid singlet-triplet qubits enable fault-tolerant single-pulse CNOT gate”	
	American Conference on Theoretical Chemistry, Chapel Hill, North Carolina	2024
	Poster: “Analog Simulation of Open Quantum Dynamics”	
	Gray-Hill lecture at the Occidental college, Los Angeles, California.	2023
	<u>Award talk</u> : “A window to Earth’s past with the help of theoretical chemistry”	
	Canadian Chemistry Conference and Exhibition , Calgary, Canada	2022
	Contributed talk: “Accurate quantum statistics from improved path-integrals in imaginary time”	
	Molecular Science Mini-meeting , Montreal, Canada	2022
	Poster: “Dimension-free ring-polymer molecular dynamics”	
	ACS Spring meeting , San Diego, California	2022
	Poster: “Accurate quantum statistics from improved path-integrals in imaginary time”	
	Geological and Planetary Sciences seminar at Caltech, Pasadena, California	2022
	<u>Invited talk</u> : “ D and ^{13}C exchange equilibria using Path-Integral Monte-Carlo”	
	Berkeley Statistical Mechanics Meeting	2020
	Poster: “Cayley modification for strongly stable path-integral molecular dynamics”	
	CECAM BioMolecular Electronics Conference , Madrid, Spain	2018
	Poster: “Principles of Charge Transport in DNA: from extensive simulations to neural networks”	
	28 th Canadian Symposium on Theoretical and Computational Chemistry, Windsor, Canada	2018
	<u>Poster prize</u> : “Charge transport in DNA: From comprehensive simulations to key principles”	
	100 th Canadian Chemistry Conference , Toronto, Canada	2017
	<u>Poster prize</u> : “Tunneling to Hopping Crossover in Thermopower of DNA Molecular Junctions”	

	Chemical Biophysics Symposium , Toronto, Canada	2017 ⁴
	Contributed talk: “DNA Molecular Junctions: Tunneling to Hopping Crossover”	
	33rd Symposium on Chemical Physics, Waterloo, Canada	2017
	Contributed talk: “Probing mechanisms of charge transport in DNA with Landauer-Büttiker formalism”	
	45th Southern Ontario Undergraduate Student Chemistry Conference , Toronto, Canada	2017
	<u>1st prize talk</u> : “Tunneling to Hopping Crossover in DNA & DNA-like molecular junctions”	
SERVICE	Departmental Graduate Studies Committee (DGSC)	2026–present
	Student monitoring committees (3 students)	2026–present
	Student Representative at the Chemistry Department Advisory Committee	2016–2017
	2 nd year representative at the Chemistry student union	2016–2017
	Board member of the <i>Chemistry Connections</i> student group	2015–2016
MEMBERSHIP	Canadian Society for Chemistry (CSC) Physical, Theoretical and Computational Chemistry (PTC) Division of the CSC Canadian Association of Theoretical Chemists (CATC) Quantum Institute at the Sherbrooke University Centre for Research in Molecular Modeling (CERMM) Centre for the Study of Advanced Materials for Batteries and Renewable Energy Storage Systems of the Sherbrooke University (CÉMAPROVUS)	
COMMUNITY VOLUNTEER INITIATIVES	Spearheaded and coordinated humanitarian supplies shipment to Ukraine 🇺🇦 from Caltech campus and beyond	2022–2023
	Volunteer at the Nova Ukraine non-profit, Stanford, California Website development, established and coordinated partnership with <i>Teach for Ukraine</i>	2021–2023
	Volunteer at the Teach for Ukraine nonprofit, Kyiv, Ukraine Recruited and interviewed candidate teachers at the remote interview stage	2021–2022
	International student orientation leader	2019, 2020
	“Big sibling” mentor for the incoming graduate students at Caltech	2019, 2020
	Science outreach program volunteer through Caltech Y	2018–2020
	High-school tutoring with CAUSE Tutoring non-profit	2018–2019
	University of Toronto Peer Tutoring group tutor	2015–2018
TEACHING	<i>Course development</i> : Computational chemistry labs, Chem3 at Caltech Focus on structure-function relations and the dangers of approximations.	2022
	High School Teacher, Rotman Arts and Science School, Vaughan, Canada Academic stream, grade 11 and 12 Chemistry, Grade 10 Science. Student placed 3rd in Vaughan, top 200 in Canada at the Avogadro chemistry contest	2020–2022
	International Chemistry Olympiad Coach Canadian National Team (4 students) – 2 bronze, 1 silver medals Ukrainian National team (1 student) – bronze medal	Spring 2018 Spring 2014
	Private tutoring of Chemistry, Physics and Math High school students: accepted to University college (UK), Columbia University (USA) and others	2014–2018
	Chemical Biology summer school, Lutsk, Ukraine Designed problems and experiments to help high school students master key concepts in chemistry	Summer 2015
PEER REVIEW	ACS journals: ACS Physical Chemistry Au APS journals: Physical Review Letters, Physical Review A, B, E and Research Elsevier journals: Organic Geochemistry, Chemical Geology	

SUMMER SCHOOLS AND WORKSHOPS	Condensed Phase Dynamics Workshop at TSRC (Virtual)	2020
	Theoretical Chemistry School at TSRC , Telluride, Colorado	2019
	Weizmann Institute of Science, Rehovot, Israel Kupcinet-Getz Scholar at Rubtchinski lab	2015
PAST EMPLOYMENT	High School Teacher, Rotman Arts and Science School, Vaughan, Canada	2020–2022
	Research Assistant, Department of Linguistics, University of Toronto Heritage language variation and change project	2016–2018
EXTRA CURRICULARS	Rock climbing, weightlifting	since 2014
	Guitar, base	since 2012
	Ukrainian folk dance	since 2004
LANGUAGES	Fluent in Ukrainian, English & Russian	
COMPUTER LANGUAGES	Python, C++, MATLAB, FORTRAN, Mathematica, Bash; Web development (PHP & django, HTML, CSS, JS)	